
ME 604: Experimental Methods in Thermal Engineering

Course Code	ME 604
Course Name	Experimental Methods in Thermal Engineering
L-T-P-C	3-0-0-3
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Fundamentals**

Importance of measurement and experimentation, calibration, uncertainty analysis, error propagation, Gaussian or Normal distribution, confidence level, regression analysis, correlation coefficient, Chi-Square test, zeroth-, first- and second-order systems. [Lectures]

- **Pressure Measurement**

Manometers, bourdon tube pressure gage, diaphragm gage, bellows gage, McLeod gage, Pirani gage and ionization gage. [Lectures]

- **Flow measurement**

Positive displacement flow meters, venturi, orifice, impact tube, flow nozzle, sonic nozzle, rotameter, pitot static tube, hot-wire anemometer, laser Doppler anemometer, flow visualization techniques – shadowgraph, Schlieren and interferometer. [Lectures]

- **Temperature measurement**

Hg-in-glass thermometer, RTD, thermistor, thermocouple, thermopile, liquid-crystal thermography, optical pyrometer. [Lectures]

- **Thermal conductivity measurement**

Guarded hot plate apparatus, heat flux meter. [Lectures]

- **Data acquisition and processing**

Signal conditioning, data transmission, storage, A to D and D to A conversion. [Lectures]

- **Designing experiments**

Text and Reference Books

1. J. P. Holman, **Experimental Methods for Engineers**, 7th edition, Tata McGraw-Hill 2001.
2. T.G. Beckwith, J.H. Lienhard V, R. D. Marangoni, **Mechanical Measurements**, 5th Edition, Pearson Education, 2010.
3. E.O. Doebelin, **Measurement systems, Application and Design**, 5th Edition, Tata McGraw-Hill, 2008